Problem Statement Lab Sheet-01
NoSQL Database Design

BCA V Semester / III Years (Session 2026-2027)

Objective: To provide practical knowledge of NoSQL databases by enabling students to install, configure, and work with MongoDB, perform database and collection management, execute basic CRUD operations, design NoSQL data models, and apply NoSQL database concepts to real-world applications..

Requirements:

- MongoDB Community Edition
- MongoDB Shell (`mongosh`)
- MongoDB Compass
- Ubuntu/Linux operating system
- Internet connection for software installation
- Basic knowledge of database concepts and command-line operations

1. Create a new MongoDB database named StudentDB and verify its successful creation using the MongoDB Shell.

2. Create the collections Students, Faculty, and Courses within the StudentDB database.

3. Insert a single document into the Students collection containing student details such as Roll Number, Name, Department, Semester, and CGPA.

4. Insert multiple student documents into the Students collection using an appropriate MongoDB command.

5. Retrieve all documents stored in the Students collection and display the results.

6. Retrieve a specific student record based on the Roll Number using a suitable query.

7. Retrieve documents by applying a condition on the Department field and display the matching records.

8. Display only selected fields (such as Name and Department) from the Students collection using field projection.

9. Update the department of a specific student using the updateOne() method.

10. Modify the semester of all students belonging to a particular department using the updateMany() method.

11. Replace an existing student document with a new document using the replaceOne() method.

12. Add a new field named Email to all documents in the Students collection.

13. Remove an existing field from all documents in the collection using an appropriate update operator.

14. Delete a specific student document based on the Roll Number using the deleteOne() method.

15. Delete multiple student records that satisfy a specified condition using the deleteMany() method.

16. Count the total number of documents present in the Students collection.

17. Rename the Students collection to StudentDetails and verify the changes.

18. Create a new collection named Departments and insert sample departmental information.

19. Drop the Departments collection and verify its successful deletion.

20. Drop the StudentDB database and confirm that all collections have been removed.

21. Recreate the StudentDB database and insert sample data after deletion.

22. Perform all CRUD operations using MongoDB Compass and compare the results with the MongoDB Shell.

23. Demonstrate the use of both MongoDB Shell and MongoDB Compass for managing databases and collections.

24. Compare the CRUD operations of MongoDB with equivalent SQL operations and summarize the differences.

25. Prepare a comprehensive laboratory report containing the objectives, commands executed, outputs, screenshots, observations, and conclusion for all CRUD operations performed in this experiment.

Instructions:

- Perform each practical using MongoDB Shell or MongoDB Compass.
- Execute and verify each command carefully.
- Record commands and outputs in the lab record.
- Test practicals with different inputs.
- Take screenshots of important outputs..

Guidelines:

- Use valid MongoDB commands and syntax.
- Maintain proper database and collection names.
- Verify the output after each operation.
- Follow proper data entry and formatting practices.
- Keep the lab record neat and complete.

Submission:

- Submit the completed lab record with commands and outputs.
- Attach screenshots of practical executions.
- Submit the MongoDB database/collection details created during the lab.